22 The functions f and g are such that

$$f(x) = 5x + 3 \qquad g(x) = ax + b \qquad \text{where } a \text{ and } b \text{ are constants.}$$

$g(3) = 20$ and $f^{-1}(33) = g(1)$

Find the value of a and the value of b.

g(3) = 20 gives 3a + b = 20. (1)

1 mark
Substitute x = 3 into g

g(1) = a + b.

1 mark
Write g(1) in terms of a and b

To invert f(x) = 5x + 3, subtract 3 then divide by 5.

$f^{-1}(33) = (33 - 3) \div 5 = 6$

1 mark
Undo f to find the inverse value

Since $f^{-1}(33) = g(1)$, a + b = 6. (2)

1 mark
Form the second equation

Subtract (2) from (1): 2a = 14, so a = 7.

Substitute into (2): 7 + b = 6, so b = −1.

Check: 3 × 7 − 1 = 20 and 7 − 1 = 6.

1 mark
Both constants correct

$a =$ 7

$b =$ −1

(Total for Question 22 is 5 marks)